

Resolve PPAD-hardness conjecture for optimal payment schemes in imperfect-information games

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Payment Schemes from Limited Information with Applications in Distributed Computing*.

1 The open problem

This paper addresses an open problem posed by Nikolaj I. Schwartzbach in *Payment Schemes from Limited Information with Applications in Distributed Computing* [1] (Economics and Computation (EC), 2022), Section 4 (Computational Complexity), page 14 (arXiv:2107.08748v2), stated as Conjecture 1 immediately after Theorem 1.. The website catalogue entry is problem 25 (W4285091592_p0) of the [Open Problems in Operations Research](#) collection.

Determine whether $\text{PaymentScheme}_t^\delta$ for finite extensive-form games with imperfect information is PPAD-hard under polynomial-time reductions.

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

Status: partial progress. The literal and intended security constraints, the imperfect-information semantics, the subgame convention, the encoding, and the unboundedness defect can all be handled rigorously. The substantive yes/no PPAD-hardness question remains exactly equivalent to whether every PPAD problem has a polynomial-time selector; I do not claim that unresolved step.

2 The Literal Positive- δ Problem

Let h_H^* be the terminal reached from the root of a subgame H by the supplied pure profile s^* . For any player i , choose $C = \{i\}$ and $s_i = s_i^*$. The literal definition permits this choice, so h_H^* is C -inducible. Security then requires

$$E_i(h_H^*) \geq E_i(h_H^*) + \delta.$$

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_075107_f43659ae, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 1/3 (minor partial progress). Maintained by the AI in Mathematics & Operations Research project.

Therefore, for every game with at least one player,

$$t \geq 1, \delta > 0 \implies \text{literal infeasibility.}$$

This is a complete result for the literal definition.

Merely requiring $s_C \neq s_C^*$ does not repair the intended meaning: a pure strategy can differ only off the realized path, while a nontrivial mixed strategy can retain h_H^* in its support. The repair must be pathwise: an “actual deviation outcome” must contain a realized action different from the intended action.

3 Exact Pathwise Formulation of Actual Deviations

Fix standard extensive-form subgames: the root is a singleton information set, and no information set straddles the subtree boundary. First assume perfect recall. For a terminal j of H , define

$$D_H(j) = \left\{ p(I) : \text{the action on the } H\text{-to-}j \text{ path at } I \text{ differs from } s_{p(I)}^*(I) \right\}.$$

Perfect recall ensures that no information set is revisited along a single path.

Lemma 1 (Reachability). *A terminal j can occur with positive probability when coalition C deviates and outsiders retain s^* if and only if*

$$D_H(j) \subseteq C.$$

Proof. Necessity follows because every noncoalition player must take the intended action. Conversely, coalition members can use the pure plans prescribing the path actions. $\square$

Thus pure, mixed-over-pure, and behavioral deviations have the same reachable-terminal set. Moreover, j is an actual realized deviation precisely when

$$D_H(j) \neq \emptyset.$$

There are two coherent coalition conventions:

- If C may contain nonactive participants and security protects every $i \in C$, the row for (H, j, i) is required exactly when

$$D_H(j) \neq \emptyset, \quad |D_H(j) \cup \{i\}| \leq t. \tag{1}$$

- If every member of C must personally change a realized action, then necessarily $C = D_H(j)$, so the row condition becomes

$$i \in D_H(j), \quad |D_H(j)| \leq t. \tag{2}$$

For the literal definition, use (1) without the condition $D_H(j) \neq \emptyset$. The honest leaf then generates the positive- δ contradiction above.

For imperfect recall, the ambiguity can also be resolved explicitly. Let $R_{Hj}(I)$ be the set of actions demanded at repeated occurrences of I .

- For pure plans or mixtures over pure plans, additionally require

$$|R_{Hj}(I)| = 1 \quad \text{for every encountered } I. \quad (3)$$

- For behavioral strategies rerandomized independently at each visit, condition (3) is unnecessary: assigning positive probability to every action in $R_{Hj}(I)$ suffices.

Both are polynomial-time path tests. Thus imperfect recall changes which rows occur but does not introduce an equilibrium or fixed-point computation.

4 An Exact Polynomial-Size LP

Since the source game has no chance nodes, let h_H^* denote the unique intended terminal of H . The security inequality for an eligible triple (H, j, i) is

$$U_i(h_H^*) - \lambda_i^\top \phi_{h_H^*} \geq U_i(j) - \lambda_i^\top \phi_j + \delta,$$

or, equivalently,

$$\boxed{\lambda_i^\top (\phi_j - \phi_{h_H^*}) \geq \delta + U_i(j) - U_i(h_H^*)}. \quad (4)$$

Consequently, under any one of the semantics fixed above, the exact program is

$$\begin{aligned} \min_{\Lambda} \quad & \sum_{(i,k): c_{ik} < \infty} c_{ik} \lambda_{ik} \\ \text{s.t.} \quad & \lambda_i^\top (\phi_j - \phi_{h_H^*}) \geq \delta + U_i(j) - U_i(h_H^*) \quad (H, j, i) \text{ eligible,} \\ & \sum_i \lambda_{ik} \geq 0 \quad k = 1, \dots, s, \\ & \lambda_{ik} = 0 \quad c_{ik} = \infty. \end{aligned} \quad (5)$$

Exactness holds in both directions. Every deviation reaching j has $D_H(j) \subseteq C$, so its security inequality appears in (5). Conversely, every row under convention (1) is witnessed by $C = D_H(j) \cup \{i\}$; under convention (2), it is witnessed by $C = D_H(j)$.

An explicitly represented tree with N nodes has at most N standard subgames and at most N terminals. All eligible triples can therefore be found by polynomially many path scans. Program (5) has ns variables and polynomially many rows.

This complexity assertion requires binary-encoded rational U , ϕ , c , and δ , with ∞ represented by a flag. For unencoded arbitrary real numbers, there is no finite-string search relation to which the ordinary class PPAD applies.

For rational data, exact LP algorithms classify (5) as infeasible, feasible with a finite optimum, or unbounded below. In the finite case they produce an exact rational optimizer of polynomial bit length; polynomial-bit Farkas and recession certificates exist in the other two cases.

5 Unboundedness Is Structural

Suppose that all c_{ik} are finite and that every ϕ_j is a probability vector. Let $\mathbf{1}$ denote the all-ones signal vector, and take any $\alpha \in \mathbb{R}^n$ satisfying $\sum_i \alpha_i = 0$. The transformation

$$\lambda_i \mapsto \lambda_i + r\alpha_i \mathbf{1} \quad (6)$$

preserves every security row, because

$$\mathbf{1}^\top(\phi_j - \phi_{h_H^*}) = 1 - 1 = 0,$$

and preserves every budget constraint, because $\sum_i \alpha_i = 0$.

Writing $C_i = \sum_k c_{ik}$, the objective changes by

$$r \sum_i \alpha_i C_i.$$

Hence, if the C_i are not all equal, every feasible instance is unbounded below: choose a zero-sum α with $\alpha^\top C < 0$.

A concrete positive- δ example exists even under the actual-deviation interpretation. Player 1 chooses L/R ; then player 2 chooses A/B at one information set containing both the post- L and post- R nodes. Let

$$s^* = (L, A), \quad t = 1, \quad \delta = 1,$$

and use one deterministic signal, with $\phi_j = 1$ at every leaf. Set

	LA	LB	RA	RB
U_1	1	0	0	0
U_2	1	0	1	0.

The only whole-game actual-deviation rows are LA versus RA for player 1 and LA versus LB for player 2; both have margin exactly 1.

Let $c_1 = 1$ and $c_2 = 2$, and set

$$\lambda_1(M) = M, \quad \lambda_2(M) = -M.$$

Then the budget sum is zero, all payoff differences are unchanged, and

$$c^\top \lambda(M) = M - 2M = -M \longrightarrow -\infty.$$

Thus the intended positive- δ formulation can be feasible yet unbounded. With $\delta = 0$, the same instance is also a feasible-unbounded instance of the literal formulation.

6 Exact PPAD Status

Let S_{can} be the rational relation whose valid outputs are:

- INFEASIBLE if (5) is infeasible;
- an exact optimizer if the optimum is finite.

On a feasible-unbounded instance, $S_{\text{can}}(I) = \emptyset$. Therefore

S_{can} is not total, and hence lies in neither TFNP nor PPAD.

This rules out PPAD-completeness, but it does *not* unconditionally rule out PPAD-hardness: partial problems can be hard.

Let S_{tot} add the output **UNBOUNDED**, and let the promise version of S_{can} require the instance to be infeasible or bounded. Both have polynomial-time selectors.

Under a domain-preserving Levin reduction, $R \leq S$ requires that

$$f(x) \in \text{dom}(S)$$

and that every $y \in S(f(x))$ decode to a solution of $R(x)$. Under this nonvacuous convention,

$$\boxed{S_{\text{tot}} \text{ is PPAD-hard} \iff S_{\text{can}} \text{ is promise-PPAD-hard} \iff \text{ENDOFLINE} \in \text{FP}.} \quad (7)$$

For the forward direction, compose a purported reduction with the polynomial-time LP selector. This yields a polynomial-time selector for **ENDOFLINE**, and hence for every PPAD relation.

Conversely, if every PPAD relation has an FP selector, map every source instance to a fixed bounded payment instance—for example, a one-player, one-terminal, one-signal instance with $\delta = 0$ and $c = 1$, whose optimizer is $\lambda = 0$. The decoder ignores the payment solution and runs the assumed source selector.

If domain preservation is omitted, the answer becomes vacuously “yes”: map every source instance to the feasible-unbounded example. Since the target fiber is empty, “every target solution decodes correctly” is vacuously true. Such a definition would make the problem hard for every search class and is therefore not meaningful. If “PPAD-hard” is defined only for total search problems, the phrase is simply inapplicable to S_{can} .

Rejection Issues

- **PPAD-hardness.** This remains open under a meaningful reduction, with the exact remaining gap being $\text{ENDOFLINE} \in \text{FP}$. No unconditional non-hardness claim is justified.
- **Actual-deviation interpretation.** Fully repaired by the pathwise condition $D_H(j) \neq \emptyset$; it still yields the exact LP (5).
- **Imperfect information and subgames.** Fully repaired once standard subgames and either perfect recall, mixed-over-pure imperfect recall, or behavioral imperfect recall are specified. Every version remains polynomial-size.
- **Encoding.** Fully repaired for binary-rational input. The arbitrary-real formulation has no ordinary PPAD classification.
- **Missing unbounded output.** Fully confirmed, including for feasible intended positive- δ instances.

Thus the strongest rigorous result is that every coherent, explicitly represented, rational formulation considered here is an exact polynomial-size LP, while the substantive yes/no PPAD-hardness question is equivalent to the unresolved collapse of PPAD to polynomial-time search.

References

- [1] Nikolaj I. Schwartzbach. *Payment Schemes from Limited Information with Applications in Distributed Computing*. Economics and Computation (EC), 2022. <https://doi.org/10.1145/3490486.3538342>